



6886 - Reconstructing the history of FUor outbursts by tracing jet knot proper motions

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Deep imaging in H2 jet tracing line (F212N) and a continuum filter (F250M)				
	1	Wide-field image	NIRCam Imaging	(1) IRAS-16289-4449

ABSTRACT

V346 Nor is the first member of the FUor class that has been observed to switch to a very low-accretion phase, followed by the onset of a new accretion outburst. VLT/SINFONI imaging reveals a complex system of H2 jet knots, whose spacing provides a record of at least 3 eruption events that took place during the past centuries. The orientation of the jet axis seems to change systematically, which suggests that the jet-launching disk is precessing. Our goals are (a) to obtain deep images of the bow-shock morphology in order to validate the precessing jet interpretation, (b) to associate, for the first time, a jet knot to a specific accretion outburst event, (c) to reconstruct, for the first time, the past outburst history of an FUor, (d) to search for the putative companion that could be responsible for triggering the outburst and the observed jet precession, and (e) to search for bodies that might have been ejected as part of the outburst-triggering event. This will provide much-needed insights on the outburst-triggering mechanism in FUors. Also, we will search for ejecta that might have been launched during the new high accretion phase event that started in 2011.

OBSERVING DESCRIPTION

JWST Proposal 6886 (Created: Thursday, October 9, 2025, 8:00:10AM Eastern Standard Time) - Overview

We propose deep JWST/NIRCam imaging in the H₂ 2.12 micrometer shock-tracer line (SW F212N filter) and the nearby continuum (LW F250M filter) that will allow us to:

- (a) Confirm the precessing jet interpretation, based on deep imaging of the bow-shock morphology;
- (b) Measure the H₂ jet knot proper motions to determine their launching dates;
- (c) Search for a jet knot that might have been launched during the 2011 outburst event;
- (d) Search for the putative companion,
- (e) Search for potential runaway bodies

Proposal 6886 - Targets - Reconstructing the history of FUor outbursts by tracing jet knot proper motions

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	IRAS-16289-4449	RA: 16 32 32.1926 (248.1341358d) Dec: -44 55 30.66 (-44.92518d) Equinox: J2000	Epoch of Position: 2000	
	Comments: IRAS16289-4449 is V346 Nor; Need for JWST: Many attempts have been made to obtain follow-up imaging of the V346\,Nor jet knots from the ground, including with the VLT instrument SINFONI, ERIS, GRAVITY, and MATISSE. However, neither of these attempts has been successful, as the visual counterpart of V346 Nor has faded since 2007, and there are no visible guidestars nearby that can be used for Coude guiding and AO correction. Therefore, it has not been possible to observe the source again with state-of-the-art ground-based high-angular resolution imaging and JWST imaging is essential. In addition, JWST is also essential for sensitivity considerations to detect fine filamentary structure, as outlined in the science case document. Category=Star Description=[Eruptive variables, FU Orionis stars] Extended=YES				

Proposal 6886 - Observation 1 - Reconstructing the history of FUor outbursts by tracing jet knot proper motions

Observation	Proposal 6886, Observation 1: Wide-field image									Thu Oct 09 13:00:10 GMT 2025
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
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Template	Module		Subarray				Target Placement			
	ALL		FULL				Module Gap			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	FULLBOX		6TIGHT		STANDARD				2
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F212N	F250M	SHALLOW4	6	1	12	12	3736.396	178351

Proposal 6886 - Observation 1 - Reconstructing the history of FUor outbursts by tracing jet knot proper motions

Special Requirements	Aperture PA Range 66 to 77 Degrees (V3 66.07457694 to 77.07457694)
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